

Coupled 0-D/1-D Fontan Model Technical Reference

Standardized model definition, equations, segments, and free parameters

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1 Coupled 0-D/1-D Fontan Model Technical Reference

This document is generated from repository sources by `scripts/docs/build_model_reference_pdfs.py`. Edit the model config, implementation notes, schematic, or this generator, then regenerate the markdown and PDF together.

1.1 Model Construction

1.1.1 Scope and Status

Reserved future model family; no executable coupled model is accepted yet.

This model family is a reserved placeholder for a future coupled 0-D/1-D closed-loop Fontan model.

No coupled baseline config, coupling interface, 1-D solver, or reference output is accepted at this stage.

This technical reference records the documentation standard that future coupled-model work must satisfy before the model can be treated as created and executable.

1.1.2 Schematic

1.1.3 Accepted Components

- planned 0-D heart, atrium, systemic beds, pulmonary beds, and fenestration
- planned true 1-D aortic subdomain
- planned true 1-D TCPC/Fontan subdomain
- planned coupling interfaces between 0-D and 1-D domains

Coupled 0-D/1-D Fontan schematic

Placeholder for the future coupled model with 1-D aorta and TCPC subdomains.

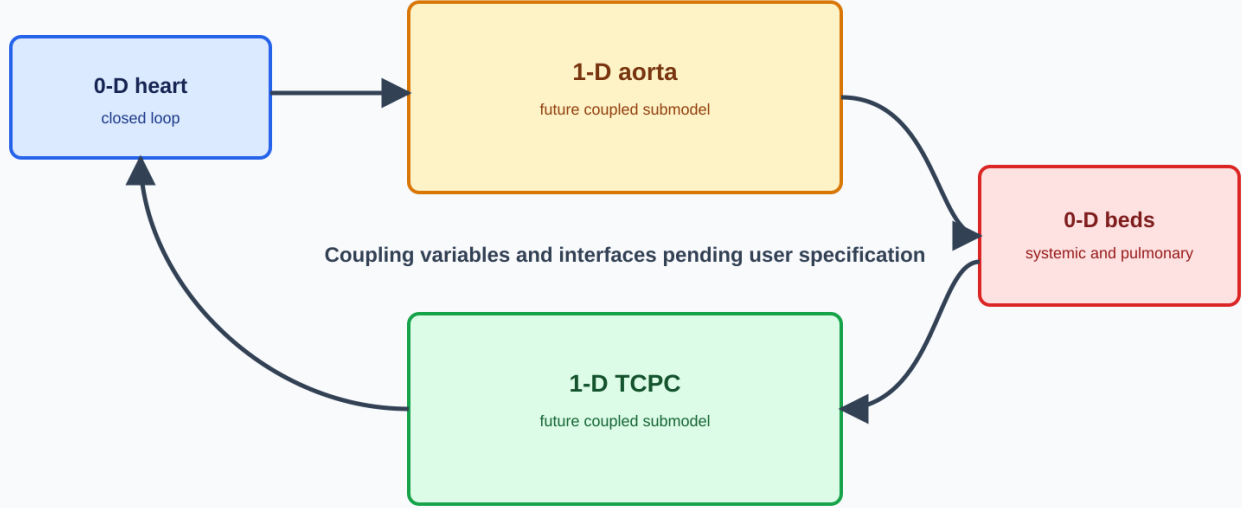


Figure 1: coupled_0d_1d schematic

1.1.4 Executable Config

No executable baseline config is accepted for this model family yet.

1.2 Governing Equations

The sign convention follows PhysioBlocks local block fluxes. For a two-node flow element, local node 1 is the first node listed in the config and local node 2 is the second node listed in the config.

1.2.1 Nodal Conservation

At every pressure node, the algebraic/differential network residual is the sum of all block flux contributions attached to that node:

$$\sum_{b \in \mathcal{B}(i)} Q_{b,i} = 0.$$

Storage blocks contribute pressure derivatives to this same residual, so closed-loop volume conservation is enforced through the connected block equations rather than through prescribed boundary flow.

1.2.2 Passive Compliance Block

For a `c_block` at pressure node `P` with capacitance `C`:

$$Q = -C \frac{dP}{dt}.$$

The saved stored volume is proportional to pressure in the local linear compliance approximation:

$$V = CP.$$

1.2.3 Pure RC Resistor Convention

This repository uses `rc_block` with zero capacitance as a pure resistive link. PhysioBlocks defines the local fluxes as:

$$Q_1 = \frac{P_2 - P_1}{R},$$

$$Q_2 = \frac{P_1 - P_2}{R} - C \frac{dP_2}{dt}.$$

When $C = 0$, the block is used as a pure resistor. To represent an upstream-to-downstream path with positive physical flow from upstream to downstream, the configs assign local node 2 to the upstream pressure and local node 1 to the downstream pressure.

1.2.4 Hydraulic R-L Link

The local quasi-vessel R-L element uses positive internal flow from local node 1 to local node 2:

$$L \frac{dQ}{dt} + RQ = P_1 - P_2,$$

$$Q_1 = -Q, \quad Q_2 = Q.$$

This is the repeated segment equation used by the quasi 0-D/1-D chains.

1.2.5 Valve R-L Block

The R-L valve block has local positive flow from node 1 to node 2 and switches conductance according to flow direction:

$$L \frac{dQ}{dt} + P_2 - P_1 + R(Q)Q = 0,$$

$$R(Q) = \begin{cases} 1/G_f, & Q > 0, \\ 1/G_b, & Q < 0, \end{cases}$$

$$Q_1 = -Q, \quad Q_2 = Q.$$

G_f is the forward conductance and G_b is the backward conductance.

1.2.6 Pulmonary RCR Windkessel

For an `rcr_block` with inlet pressure P_1 , outlet pressure P_2 , middle pressure P_m , proximal resistance R_1 , distal resistance R_2 , and compliance C :

$$Q_1 = \frac{P_m - P_1}{R_1},$$

$$Q_2 = \frac{P_m - P_2}{R_2},$$

$$\frac{P_1 - P_m}{R_1} + \frac{P_2 - P_m}{R_2} - C \frac{dP_m}{dt} = 0.$$

1.2.7 Active Atrium

The active atrium is a one-node time-varying elastance chamber:

$$E(t) = E_{min} + (E_{max} - E_{min})a(t),$$

$$V_a(t) = V_{0,a} + \frac{P_a(t) - P_{ext}}{E(t)},$$

$$Q_a = -\frac{dV_a}{dt}.$$

The activation $a(t)$ is a raised-cosine pulse over the configured start, peak, and end phase of the cardiac cycle.

1.2.8 Spherical Ventricular Cavity

The active ventricular cavity stores volume through the spherical cavity displacement y , reference radius R_0 , and wall thickness d_0 :

$$V(y) = \frac{4\pi}{3} \left[R_0 \left(1 + \frac{y}{R_0} \right) - \frac{d_0}{2 \left(1 + \frac{y}{R_0} \right)^2} \right]^3,$$

$$Q_v = -\frac{dV(y)}{dt}.$$

The pressure-displacement relation comes from the configured PhysioBlocks spherical dynamics, velocity law, passive rheology, and active macro-Huxley submodels. The corresponding free parameters are listed in the parameter inventory below with the `cavity.*` prefix.

1.3 Segment Inventory

No executable segment inventory exists yet. The first coupled model implementation must list every 0-D block, every 1-D segment, every coupling interface, and every parameter source in this section before acceptance.

1.4 Free Parameters

Unless a parameter states otherwise in the config, units follow the repository SI convention: pressure in Pa, flow in $\text{m}^3 \text{s}^{-1}$, resistance in Pa s m^{-3} , capacitance in $\text{m}^3 \text{Pa}^{-1}$, inertance in $\text{Pa s}^2 \text{m}^{-3}$, volume in m^3 , length in m, and time in s.

No executable free-parameter set is accepted yet for this model family. The future coupled implementation must list every 0-D parameter, every 1-D material/geometric parameter, every coupling-interface parameter, and every calibration bound here.

1.5 Documentation and Regeneration

Model-local documentation artifacts:

- `models/coupled_0d_1d/README.md`
- `models/coupled_0d_1d/docs/coupled_0d_1d_schematic.svg`
- `models/coupled_0d_1d/docs/coupled_0d_1d_schematic.png`
- `models/coupled_0d_1d/docs/implementation_notes.md`
- `models/coupled_0d_1d/docs/coupled_0d_1d_technical_reference.md`
- `models/coupled_0d_1d/docs/coupled_0d_1d_technical_reference.pdf`

Regenerate the technical reference source and PDF with:

```
python3 scripts/docs/build_model_reference_pdfs.py --model coupled_0d_1d
```

1.6 Current Limitations

- No equations are instantiated in an executable coupled config yet.
- No free parameters are accepted for simulation yet.
- The equations and segment inventory below are requirements for the future implementation, not evidence of a validated coupled model.

The model parameters and standardized data are for computational development and calibration workflows. Simulation outputs must not be presented as clinically validated without separate validation and documentation.